

Extreme Programming (XP) Summary

1. What is Extreme Programming (XP)?

Extreme Programming (XP) is an Agile software development methodology aimed at **quickly delivering high-quality software** that provides the maximum value to the customer. XP emphasizes **values, principles, and practices** to ensure both speed and quality in development.

2. Core Values and Principles

XP is guided by five core values:

Communication – Encourage constant interaction between team members and customers.

Simplicity – Implement only what is necessary now, avoiding over-design.

Feedback – Use tests and customer input to continuously adapt.

Courage – Be willing to refactor, improve, and face challenges.

Respect – Foster trust and appreciation among all team members.

These values shape the mindset and behavior of the team throughout the project.

3. Core Practices

XP defines several engineering practices to achieve its goals:

Pair Programming – Two developers work together on the same code to improve quality and share knowledge.

Test-Driven Development (TDD) – Write tests before writing the actual implementation.

Continuous Integration (CI) – Integrate and test code frequently to detect issues early.

Refactoring – Continuously improve code structure without changing functionality.

Small Releases – Deliver functional software in small, frequent iterations.

User Stories – Customers describe requirements in a simple, narrative format.

4. Roles and Responsibilities

XP assigns specific roles within a development team:

Tracker – Tracks project progress and ensures tasks are on schedule.

Manager – Coordinates resources and task assignments.

Customer (Client) – Provides requirements through user stories.

Programmer – Every team member contributes to coding.

Coach – Guides the team in following XP principles and practices.

Doomsayer – Identifies potential risks and issues.

Integration Tester – Ensures proper system integration testing.

Usability Tester – Tests for user-friendliness (ideally more than one).

Report Writer – Documents development progress.

Demo & Poster – Prepares project demonstrations and presentations.

Bitbucket / XP Experts – Provide expertise in tools and methodology.

Researchers – Software Researchers and Existing System Researchers investigate improvements and solutions.

Note: *Tracker* and *Manager* are rotating leadership roles.

5. User Stories

User stories are short, simple descriptions of a feature told from the perspective of the customer.

Typical format:

“As a [user role], I want [feature], so that [value].”

Each user story is linked to an **Engineering Task** and managed as an issue ticket.

6. Engineering Tasks

Engineering tasks are the concrete development activities derived from user stories.

Each task corresponds to an issue ticket for tracking and accountability.

Tasks are designed to be small, quickly completed, and frequently delivered.

7. Conclusion

Extreme Programming (XP) is a powerful Agile methodology that focuses on **customer value, rapid delivery, continuous feedback, and team collaboration**.

By combining **values** → **practices** → **roles** → **user stories** → **engineering tasks**, XP provides a complete framework for managing software projects in fast-changing environments.